Airborne Platform-Assisted Scalable Group Quantum Key Distribution for 6G Network

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Outline

I. Introduction

II. Proposed System

III. Results

IV. Conclusion

Background: Security of 6G Networks

The Emergence of 6G: Driving the need for secure group communications

- Simultaneous data exchange among massive distributed wireless users
- Require highly secure, low-latency group communication over wide areas

The Threat of Quantum Computers: Classical cryptography is vulnerable to future quantum computer attacks.

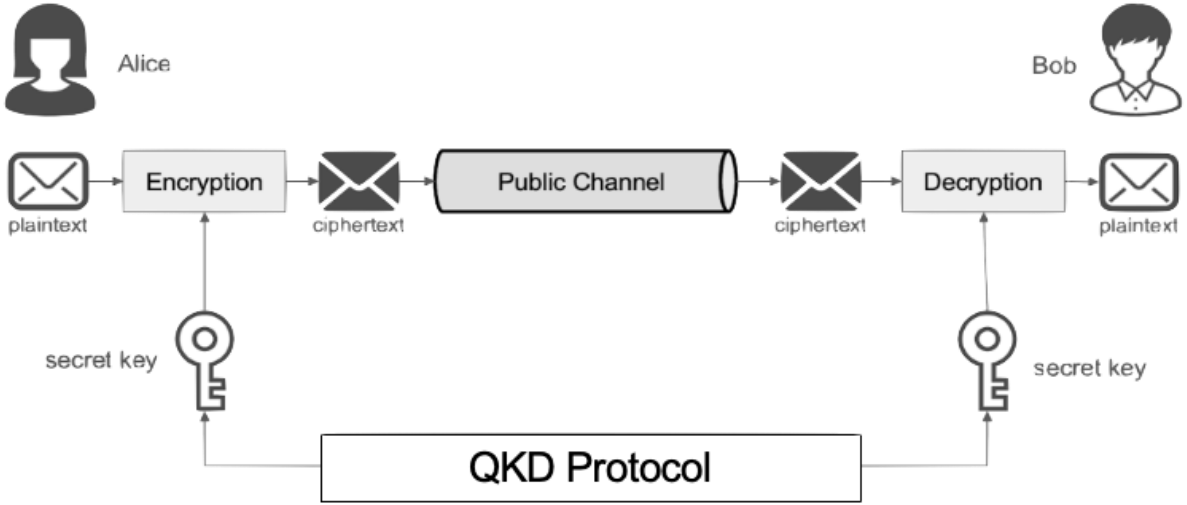
Scalable and efficient distribution of a **quantum-resistant group key** to support secure group communications.



Quantum Key Distribution (QKD)

QKD is a secure communication technology to distribute secret keys

- **Based on quantum physics**
- **Guarantees unconditional security**



Research Motivation - Limitations of Current Infrastructure

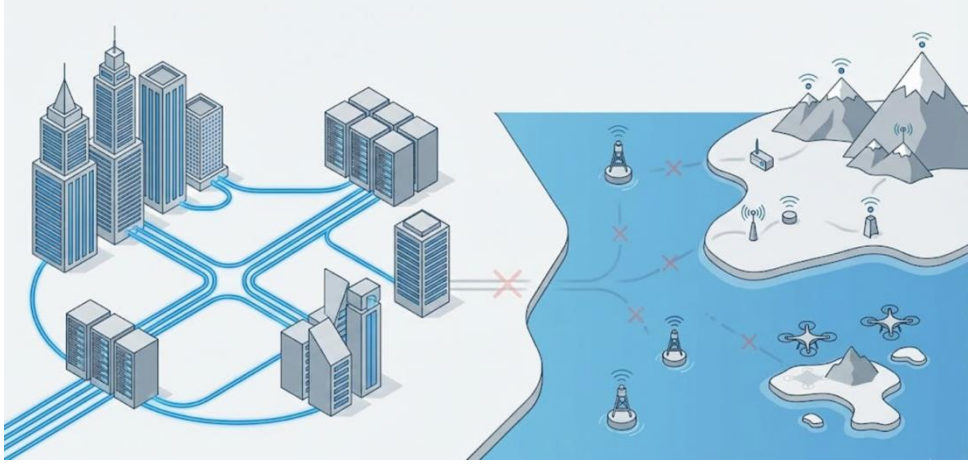
Current QKD Implementation

- Relies on **Optical Fiber** channels.
- Suitable for static networks between central servers (e.g., Banks).

Challenges in 6G Environment

- 6G devices are distributed in **remote areas** (e.g., mountains, islands).
- Laying optical fiber to every device is **costly and impractical**.

→Fiber-based QKD may not be feasible for 6G networks.



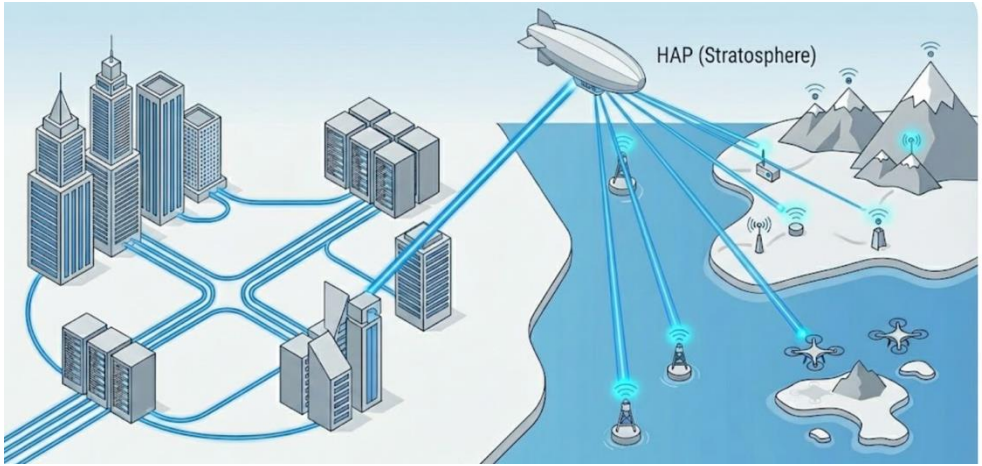
Research Motivation - Proposed Solution: HAP-based FSO

Wireless Solution: Free-Space Optics (FSO)

- Replaces physical fiber with **wireless FSO links**.

High-Altitude Platform (HAP)

- Operates in the **stratosphere** (Approximately 20km).
- Acts as a central node to cover **wide areas** and **remote regions**.
- Supports ground devices without physical fiber.



Group Quantum Key Distribution (Group QKD)

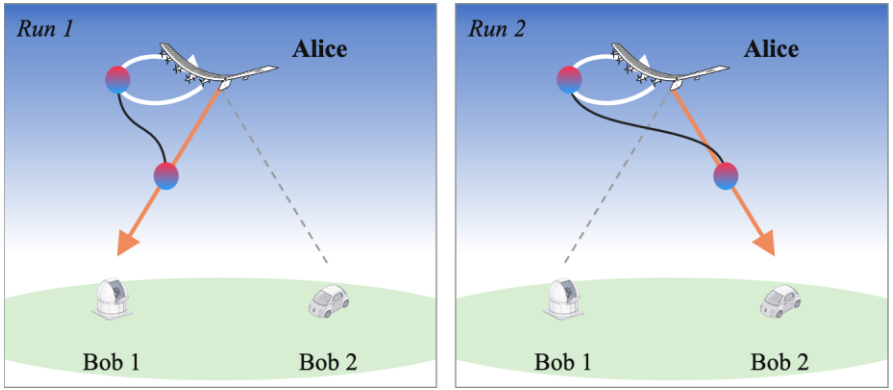
- **Group QKD**: shares the same secret key (*group key*) among N parties

Two-party QKD (e.g., BBM92)

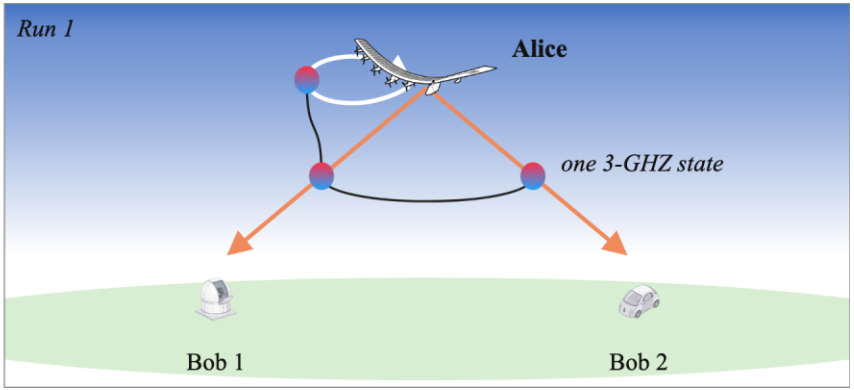
- QKD with each user, one by one ⇒ $N-1$ runs
- Then the group key is encrypted with each key

N-GHZ-QKD $|\mathrm{GHZ}_N\rangle = \left(|0\rangle^{\otimes N} + |1\rangle^{\otimes N}\right)/\sqrt{2}$

- One entangled state shared by all N parties
- Same basis (Z) ⇒ identical bits for all ⇒ 1 run



(a) Two-party QKD



(b) 3-GHZ-QKD

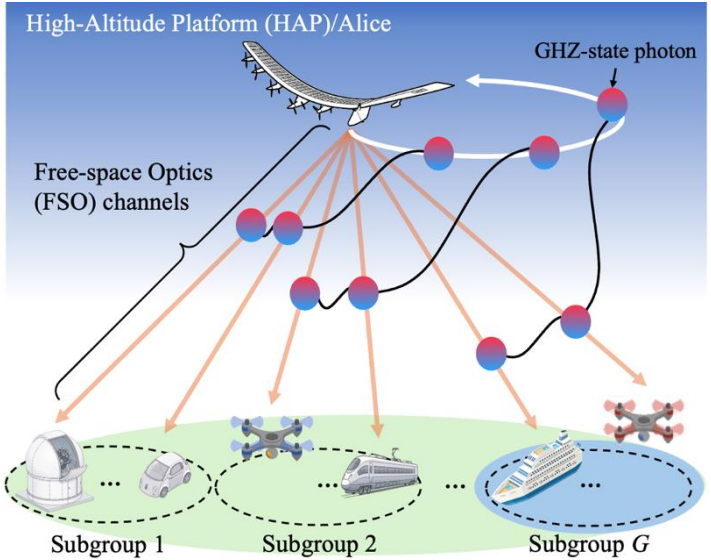
Previous Work: Single-HAP N-GHZ-QKD (APWCS 2026)

Considered scenario: one HAP distributes a group key to many ground users over FSO links

Large N-GHZ states are still at the research stage: hard to generate and distribute

- ❑ Aim: feasible GHZ states (3-, 4-GHZ) + TDMA ⇒ fewer key-distribution steps than two-party QKD

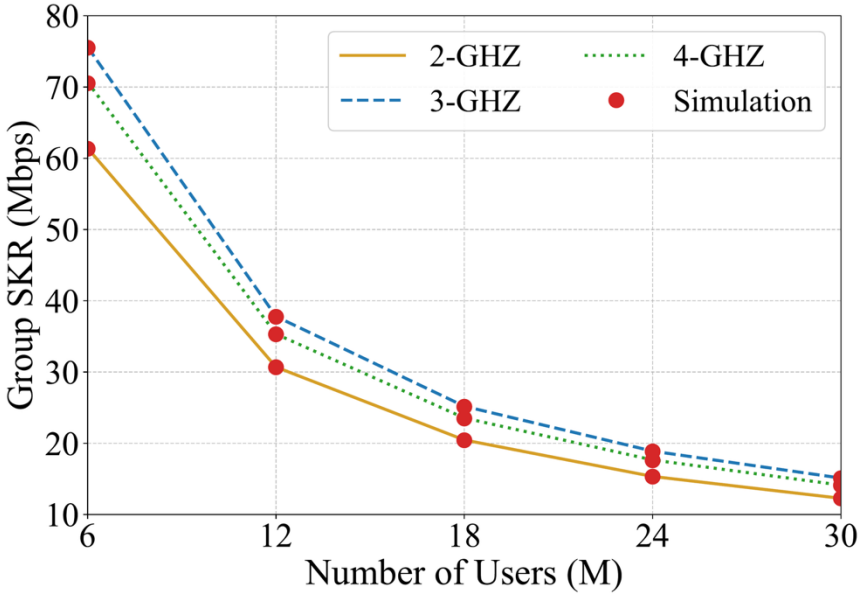
The users are divided into *subgroups*; each subgroup shares an N-GHZ state with the HAP



- ❑ **Phase 1**: subgroups are served *one by one* (TDMA) ⇒ a key for each subgroup
- ❑ **Phase 2**: the HAP sends the group key, encrypted with each subgroup key
  - ❑ All users obtain the same group key

⇒ *The HAP serves only one subgroup in each time slot*

Issue with the Previous Work: Limited Number of Users

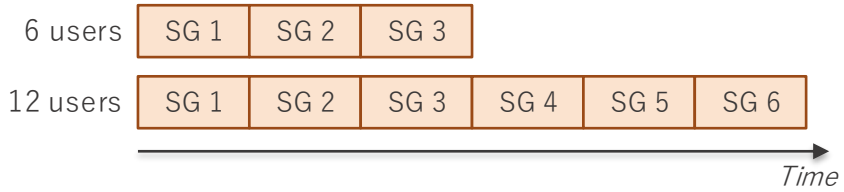


Group SKR versus number of users

- The group SKR decreases as the number of users increases
- The same trend for 2-, 3-, and 4-GHZ

Cause: TDMA

- One HAP serves only one subgroup per slot

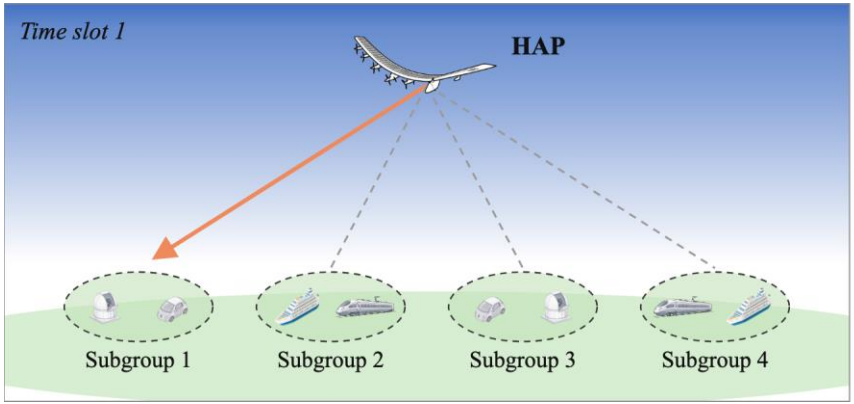


- More users → more subgroups → longer TDMA cycle → lower group SKR

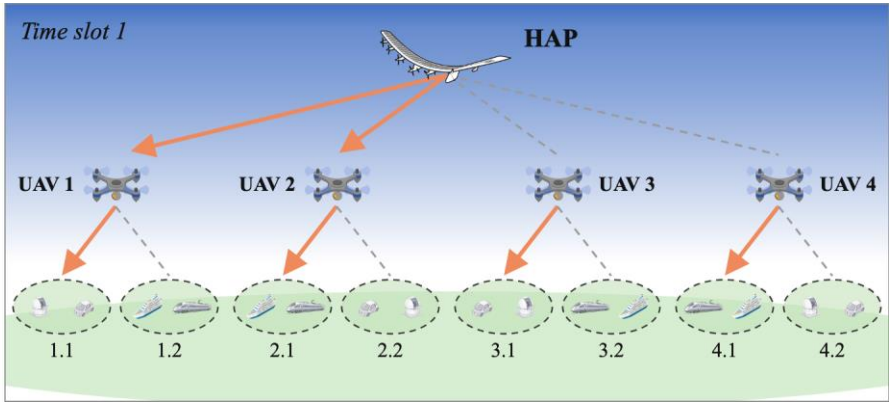
⇒ *The number of users that one HAP can support is limited*

Goals

- **Goal**: support *many more users* with one group key, at a high group SKR



(a) Previous work: 1 subgroup per time slot



(b) This work (HAP + UAVs): 4 subgroups per time slot

- UAVs serve their own subgroups *in parallel* ⇒ shorter TDMA cycle ⇒ more users
  - A HAP–UAV group QKD scheme with two-tier parallel TDMA
  - A timeline-based formulation of the group SKR
  - Evaluation versus the number of users, compared with the previous work

Outline

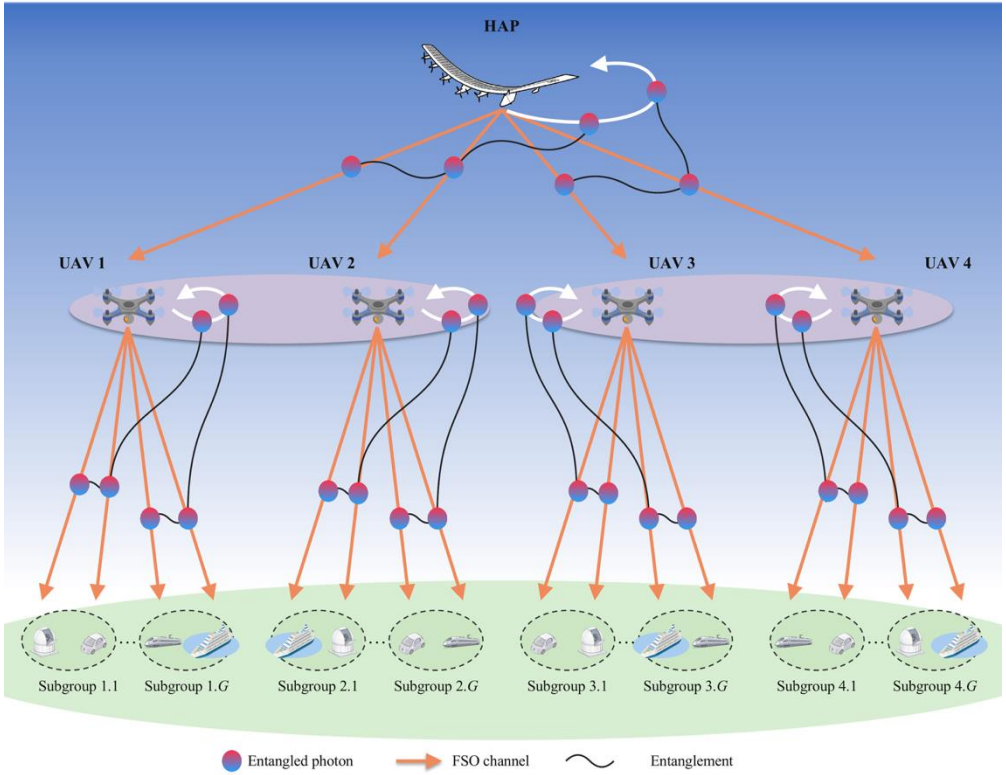
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System Model

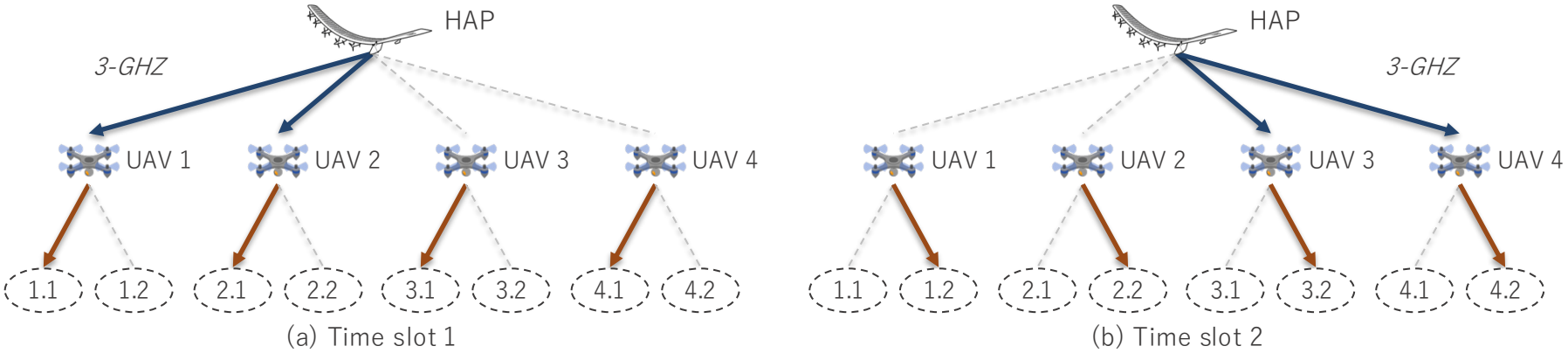


- **HAP**: generates the group key
  - 3-GHZ state with *two UAVs* at a time
- ***L* UAVs**: UAV 1, …, *L*
  - 3-GHZ state with *one subgroup* (2 users) at a time
- ***M* ground users**: divided evenly among the UAVs
  - $G = M/(2L)$ subgroups per UAV
  - Subgroup $\ell.g$: g-th subgroup of UAV ℓ

⇒ Two TDMAs run in parallel: HAP → two UAVs at a time (L/2 turns), each UAV → its subgroups (G turns)

Phase 1: Quantum Subgroup Key Distribution

Example: 4 UAVs, 16 users (2 subgroups of 2 users per UAV)

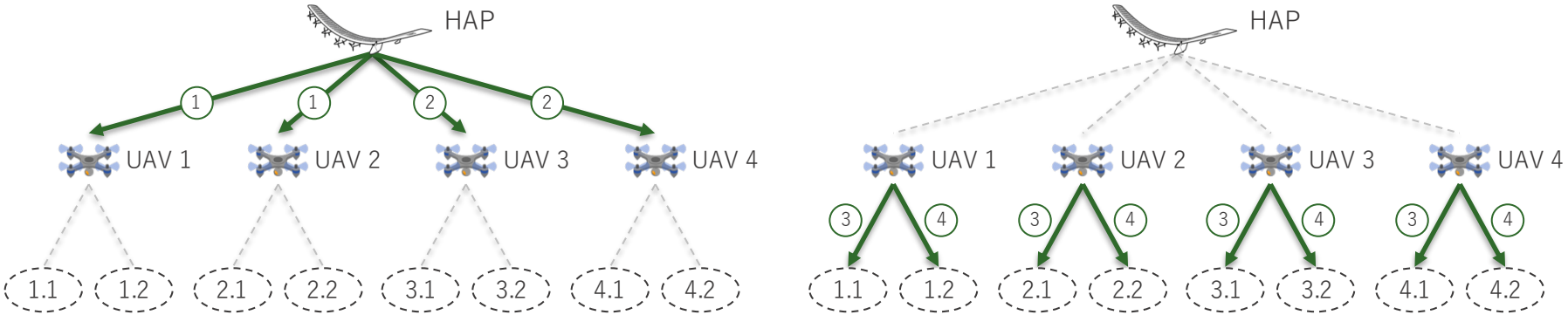


- In each time slot (Quantum phase + Post-processing), run **at the same time**
  - HAP → two UAVs with a 3-GHZ state: UAV 1 and 2, then UAV 3 and 4
  - Every UAV → one of its subgroups with a 3-GHZ state: Subgroup ℓ.1, then ℓ.2

  ⇒ *Keys after Phase 1: HAP–UAV keys and UAV–subgroup keys*

Phase 2: Group Key Distribution

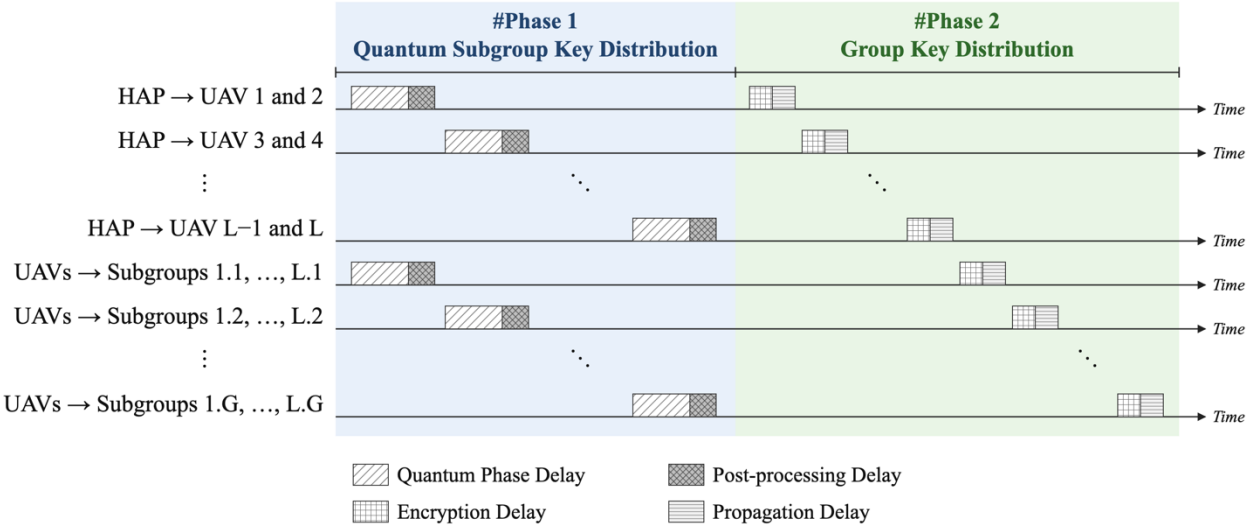
Example: 4 UAVs, 16 users (same as Phase 1)



- The HAP generates the group key and relays it in order ①–④ (one-time-pad encryption)
  - ①, ②: HAP → UAV 1 and 2, then UAV 3 and 4 (group key ⊕ HAP–UAV key)
  - ③, ④: Every UAV decrypts it → Subgroup ℓ.1, then ℓ.2 (group key ⊕ UAV–subgroup key)

  ⇒ *All 16 users obtain the same group key*

Timeline and Group Secret Key Rate (SKR)

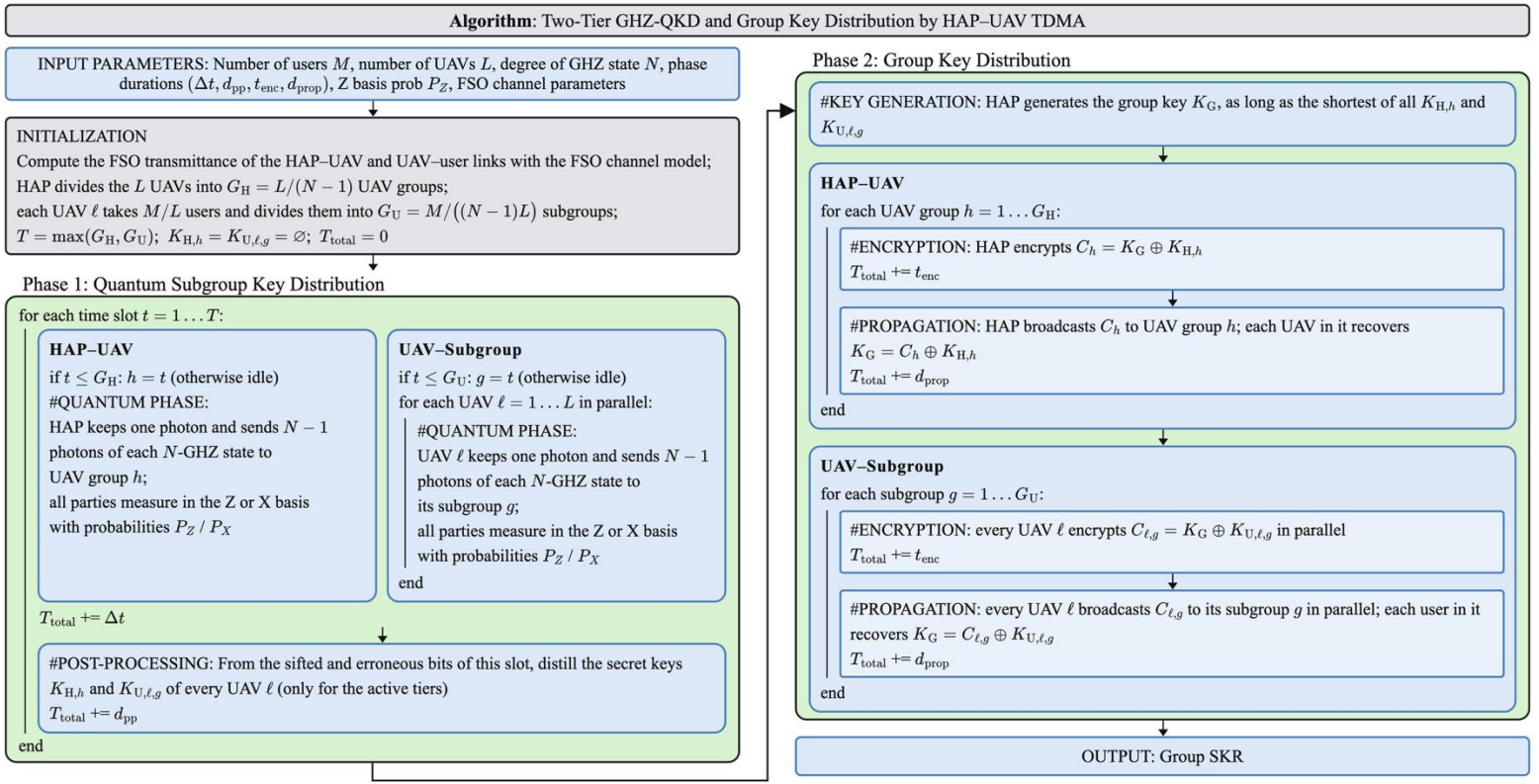


where

- AKR_H: key rate, HAP → UAVs
- AKR_U: key rate, UAV → users
- min: shorter key = group key
- Δt : quantum phase delay
- d_{pp} : post-processing delay
- t_{enc}, d_{prop}: Phase 2 delays
- T : Number of phase1 time slots
- L: Number of UAVs
- G: subgroups per UAV

$$\text{Group SKR} = \frac{\min(\text{AKR}_\text{H}, \text{AKR}_\text{U}) \cdot \Delta t}{T(\Delta t + d_{\text{pp}}) + (L/2 + G)(t_{\text{enc}} + d_{\text{prop}})}$$

Algorithm of the Proposed Scheme



Outline

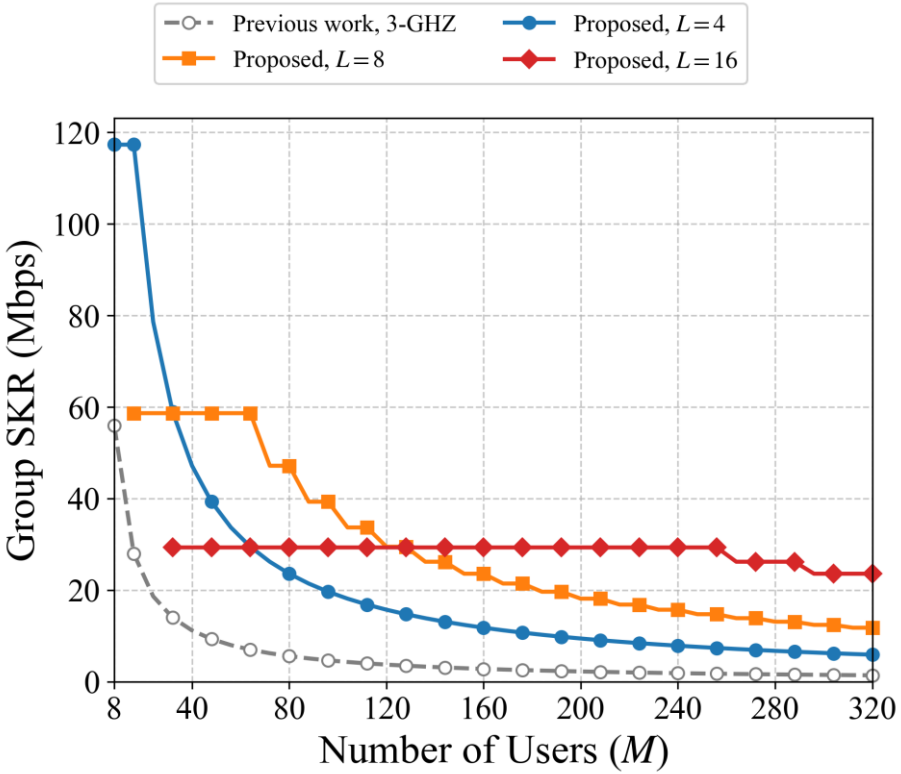
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Result: Group SKR versus Number of Users



- **Settings**: L = 4, 8, 16 UAVs, 3-GHZ
- Higher group SKR than the previous work for all numbers of users
- Few users: fewer UAVs are better
  - More UAVs ⇒ more HAP turns ($L/2$)
- Many users: more UAVs are better
  - More UAVs ⇒ fewer subgroups per UAV (G)

  ⇒ *The best number of UAVs grows with the number of users*

Conclusion

- ❑ Proposed a HAP–UAV group QKD scheme with two-tier parallel TDMA
- ❑ Formulated the group SKR based on the timeline
- ❑ The best number of UAVs grows with the number of users

Thank you

Appendix

Group Quantum Key Distribution: N-GHZ-QKD

N-GHZ (Greenberger-Horne-Zeilinger) state:

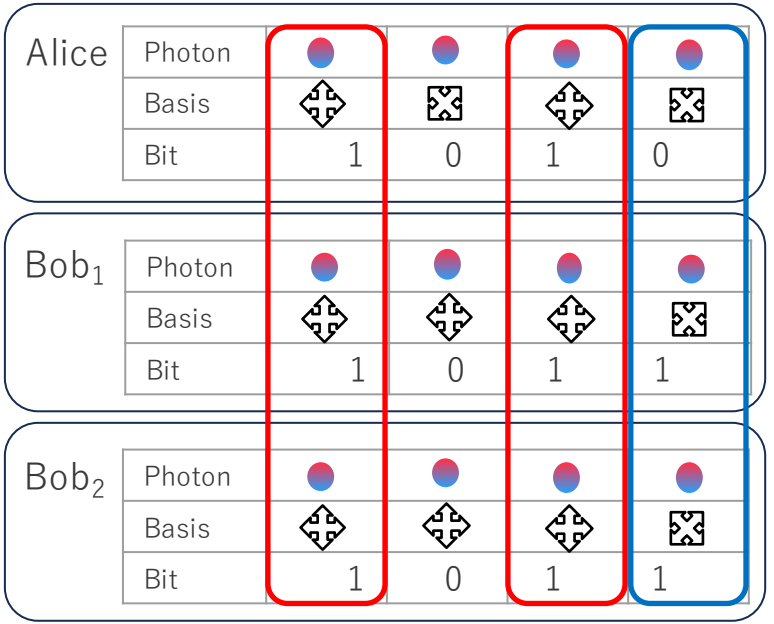
- A quantum entangled state shared among N parties
- Measuring in the same basis (Z basis) yields perfectly correlated outcomes among all parties

Benefits for group key distribution:

- A single execution establishes secret key shared by all N parties

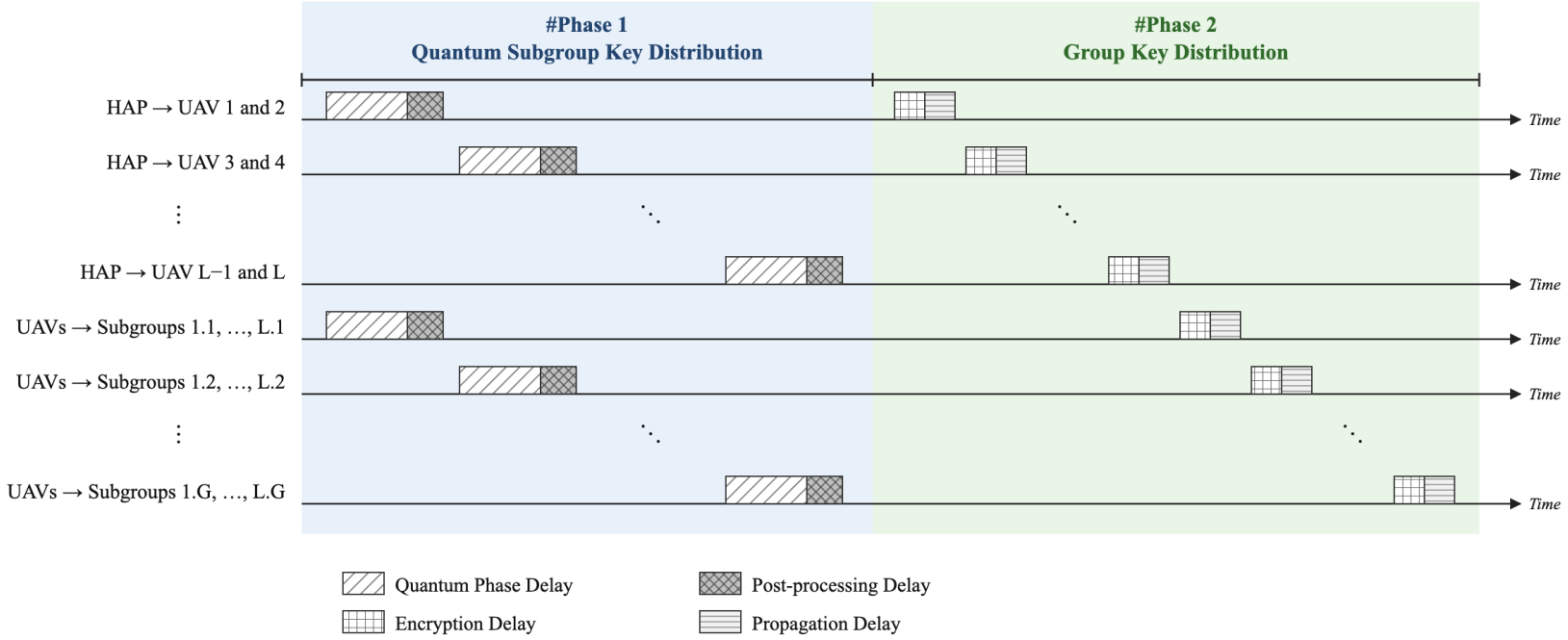
E.g. 3-GHZ-QKD

$$|GHZ_3\rangle = \frac{1}{\sqrt{2}}(|000\rangle + |111\rangle)$$

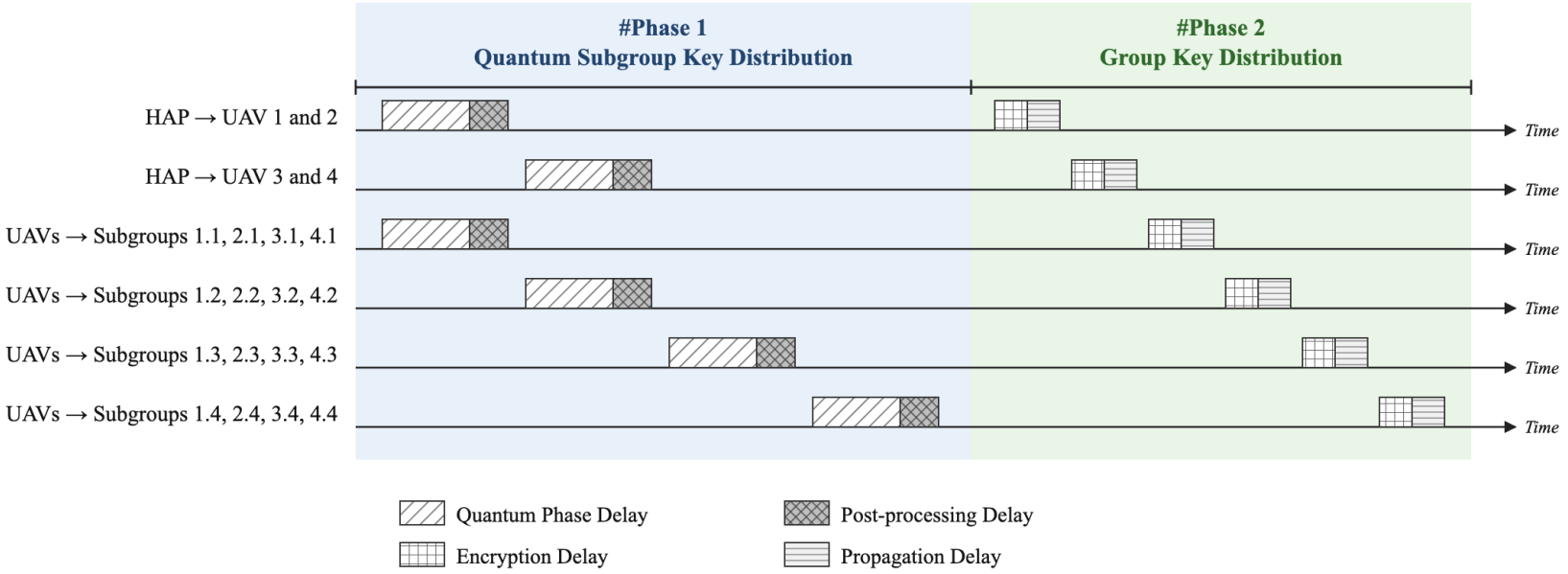


Record “11” as the sifted key

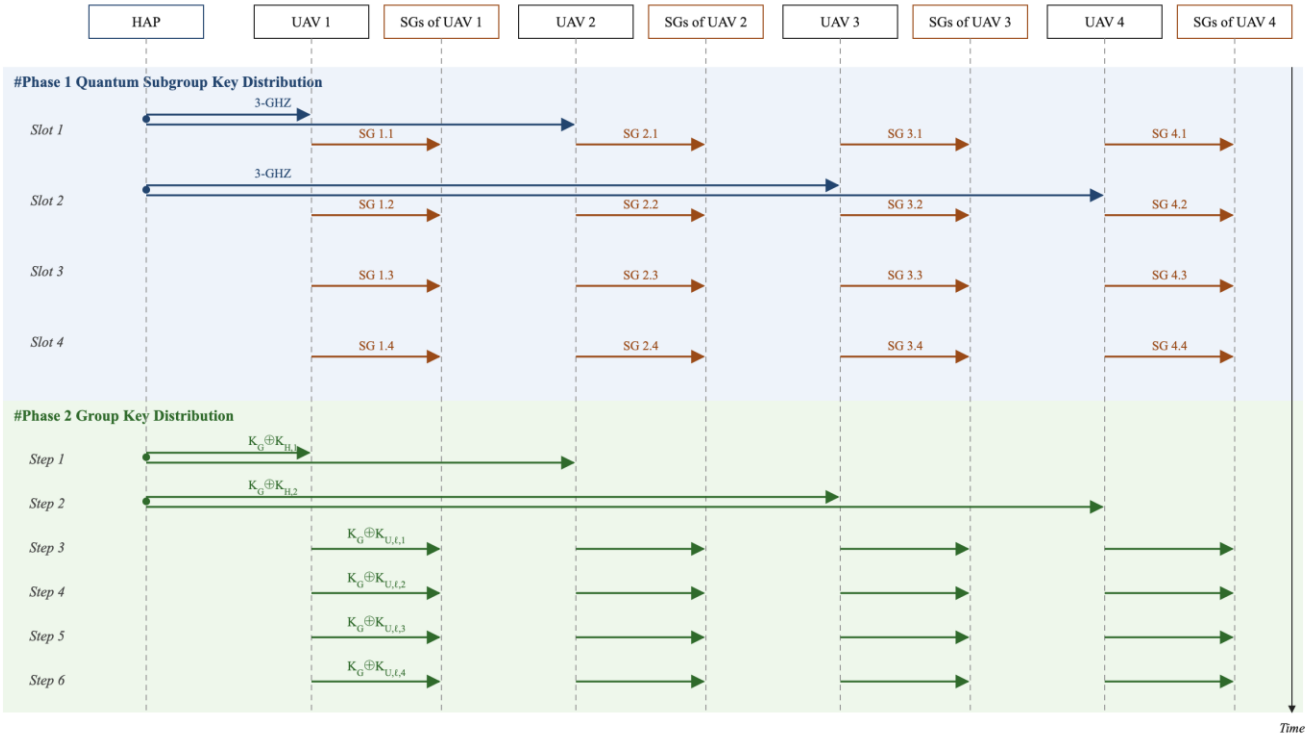
Appendix: Timeline (General Form)



Appendix: Timeline (4 UAVs, 32 Users)



Appendix: Sequence (4 UAVs, 32 Users)



NQKD(GHZ)

$$|GHZ\rangle = \frac{1}{\sqrt{2}}(|000\rangle + |111\rangle)$$

